



Ketogenic diet treatment increases longevity in *Kcna1*-null mice, a model of sudden unexpected death in epilepsy

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Epilepsia, 57(8):e178–e182, 2016
doi: 10.1111/epi.13444



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SUMMARY

Individuals with poorly controlled epilepsy have a higher risk for sudden unexpected death in epilepsy (SUDEP). With approximately one third of people with epilepsy not achieving adequate seizure control with current antiseizure drugs, there is a critical need to identify treatments that reduce risk factors for SUDEP. The *Kcna1*-null mutant mouse models risk factors and terminal events associated with SUDEP. In the current study, we demonstrate the progressive nature of epilepsy and sudden death in this model (mean age of mortality ($\pm$ SEM), postnatal day [P] 42.8 ± 1.3) and tested the hypothesis that long-term treatment with the ketogenic diet (KD) will prolong the life of *Kcna1*-null mice. We found that the KD postpones disease progression by delaying the onset of severe seizures and increases the lifespan of these mutant mice by 47%. Future studies are needed to determine the mechanisms underlying the KD effects on longevity.

KEY WORDS: Kv1.1 knockout, Mortality, Survival, Disease progression, Seizures.

Sudden unexpected death in epilepsy (SUDEP) occurs in at least 1:1,000 people with epilepsy each year.¹ This number dramatically increases to approximately 1:150 each year for people with poorly controlled and severe epilepsy.² Approximately 30% of individuals with epilepsy do not achieve adequate seizure control with current antiseizure drugs. Thus, there is a critical need to identify treatments for refractory epilepsy that increase longevity in people at risk for SUDEP.

One of the few effective, nonsurgical treatments against medically intractable epilepsy is the high-fat, low-carbohydrate/protein ketogenic diet (KD). Treatment with the KD has been reported to completely abolish seizures in up to 13% of patients and reduce seizure frequency by >50% in approximately two thirds of patients with refractory epilepsy.³ We have previously reported that the KD effectively reduces seizures in *Kcna1*-null mutant mice, which model early onset epilepsy, multiple temporal lobe epilepsy syndromes, and SUDEP.^{4–13} Due to the broad-spectrum efficacy of KD in controlling seizures, in the current study we asked whether KD treatment can increase longevity in *Kcna1*-null mice.

METHODS

Animals

Breeding pairs of heterozygous *Kcna1*-null mice on a C3HeB/FeJ congenic background were purchased from Jackson Laboratories (Bar Harbor, ME, U.S.A.), and the

Accepted May 25, 2016; Early View publication June 27, 2016.

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colony was maintained in the Animal Resource Facility at Creighton University School of Medicine. Mice were given food and water ad libitum and kept on a 12-h light/dark cycle. Tail clips were collected on postnatal day (P) 12–15, and genotypes were determined by Transnetyx, Inc (Cordova, TN, U.S.A.). All experiments conformed to National Institutes of Health (NIH) guidelines in accordance with the United States Public Health Service's Policy on Humane Care and Use of Laboratory Animals and were approved by the Institutional Animal Care and Use Committee at Creighton University School of Medicine.

Seizure ontogeny

Seizure frequency and severity of immature *Kcna1*-null mice were determined by independent direct observation of behavioral seizures by 2–5 trained scientists. Mice were observed continuously for the first 15 min of each hour from zeitgeber time (ZT) 02:00–10:00 h (lights on at ZT 00:00 h), a period of relative higher seizure frequency in this animal model.^{4,14–16} Seizure severity was scored using a modified Racine scale, as we have described previously.^{4,6,14–16} Stage 1, myoclonic jerk; stage 2, head stereotypy; stage 3, forelimb/hindlimb clonus, tail extension, a single rearing event; stage 4, continuous rearing and falling; and stage 5, severe tonic-clonic seizures. We have previously reported that stage 1 behavior is associated with an EEG spike-wave discharge and that stages 2–5 are associated with a sustained increase in EEG amplitude and time-dependent frequencies in approximately P40 *Kcna1*-null mice.^{4,5,14–16} All mice were observed in their home cage. Pregenotyped mice were identified with tail markings, and the behavior of all mice was documented. Once genotype was determined, only *Kcna1*-null mice were observed thereafter. If the view of a mouse was obstructed during an epoch, that epoch was discounted and that mouse was observed for an additional epoch.

KD treatment

Kcna1-null mice were randomly weaned onto either a standard diet (SD) or a KD (6.3:1, fat to carbohydrates plus proteins; Bio-Serv F3666, Frenchtown, NJ, U.S.A.) at approximately P21.^{4–6} Mice remained on the treatment and undisturbed for the duration of their life. Age of mortality was recorded.

Blood β -hydroxybutyrate (BHB) and glucose measurements

In a subset of mice, blood β -hydroxybutyrate (BHB) and glucose levels were measured at weaning prior to initiation of treatment to establish a baseline, and then approximately 10 days throughout treatment until natural death (*Kcna1*-null mice) or termination for age-matched wild-type (WT) controls. BHB and glucose levels were measured to verify ketosis as we and others have done from blood samples (50 μ l) collected from the tail vein using a test strip system

and reader (Precision Xtra; Abbott Diabetes Care Inc., Alameda, CA, U.S.A.).^{5,6}

Statistics

Statistical analysis was performed with Prism 6 (Graph-Pad Software, Inc., La Jolla, CA, U.S.A.). Differences across age groups and between treatments were determined using two-way analysis of variance (ANOVA) with appropriate post hoc test unless otherwise noted. Survival curves were constructed using the Kaplan-Meier method and comparisons made with the Log-rank test.

RESULTS

Seizure frequency increases with age until sudden death

Behavioral seizures were not detected in *Kcna1*-null mice until after P21. The number of seizures continued to increase with age ($F_{2,24} = 19.43$, $p < 0.0001$) (Fig. 1A). Compared to P25–29, *Kcna1*-null mice experienced significantly more seizures at P30–39 ($p < 0.0001$) and P40–49 ($p < 0.001$). Both mild (stages 1–2) and more severe phenotypes (stages 3–5) increase with age, but the distribution of phenotypes among seizures remained similar throughout life (Fig. 1B). Our previous video-electroencephalography (EEG) analyses reported similar severe seizure phenotypes in approximately P40 *Kv1.1* KO mice.^{4,5,14–16} Seizures continued until death, which typically occurred between postnatal weeks 5 and 7 in our colony with a mean age of $P42.8 \pm 1.3$ (SEM) and median age of P45 (range = P37–78; $n = 90$) (Fig. 2A,B, black line). Observed deaths always followed a severe seizure event ($n = 16$), similar to previous reports.^{7–9}

KD treatment delays seizure progression and increases longevity

Previously, we reported that KD treatment reduces seizure number in vivo and hippocampal hyperexcitability in *Kcna1*-null mice (approximately P35–45).^{4–6} Here, littermates were randomly weaned on approximately P21 onto either the SD or KD. KD treatment significantly reduced seizure number when compared to control *Kcna1*-null mice ($F_{1,24} = 19.31$, $p < 0.001$), specifically at P30–39 ($p < 0.001$) and P40–49 ($p < 0.05$, Sidak's multiple comparisons post hoc test, Fig. 1A). A significant interaction ($F_{2,24} = 4.43$, $p < 0.05$) and post hoc tests indicate that although seizure numbers were similar among the first three age groups of KD-treated mice, seizure number began to increase by P50–59 ($p < 0.005$; the control *Kcna1*-null cohort did not survive past the P40–49 time point).

Examination of the seizure profile indicates control *Kcna1*-null mice experienced more severe seizure phenotypes when compared to age-matched KD-treated mice at P25–29 (Fisher's exact test, $p < 0.0001$) and 30–39 ($p < 0.05$) (Fig. 1B). By P50–59, the seizure profile of KD-treated mice began to resemble the control *Kcna1*-null mice

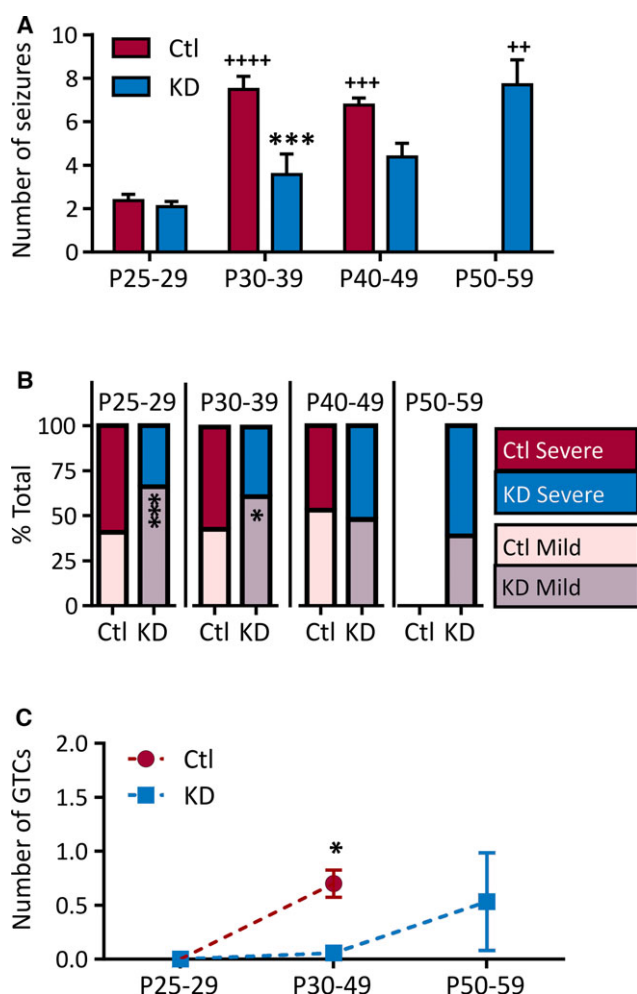


Figure 1. KD treatment delays seizure progression in *Kcna1*-null mice. (A) The increase in seizure number after P30 in control SD-treated control *Kcna1*-null mice (Ctl, red) is attenuated in KD-treated mice (KD, blue). Data are plotted as the mean $\pm$ standard error of the mean (SEM). **Differs from Ctl P25–29, $p < 0.005$, *** $p < 0.001$, **** $p < 0.0001$. ***Differs from Ctl within age group, $p < 0.001$. (B) Ratio of mild to severe seizures are plotted and analyzed within age groups, * $p < 0.05$, *** $p < 0.0001$. (C) Onset of severe generalized tonic–clonic (GTC) seizures increases prior to sudden death for both Ctl and KD-treated groups (Mann–Whitney t -test, * $p < 0.05$) ($n = 4$ –7 for each group at each time point). *Epilepsia* © ILAE

seizure profile prior to sudden death, namely, seizure number (Fig. 1A), severity (Fig. 1B), and the number of stage 5 GTC seizures (Fig. 1C). These data suggest that the KD delays disease progression in this severe epilepsy model.

KD treatment significantly increased the lifespan of *Kcna1*-null mice by approximately 47% from $P43.2 \pm 1.9$ to 63 ± 2 days ($n = 22$ Ctl and 10 KD, $p < 0.0001$) (Fig. 2A,B). Median lifespans were P46 and P62.5 for Ctl and KD-treated mice, respectively. Unfortunately, none of the deaths of KD-treated mice occurred during video recording; therefore, whether they were seizure-related is

unknown at this time. These data indicate that KD treatment reliably increases longevity of *Kcna1*-null mice when compared to SD-treated controls.

Blood BHB levels are increased and glucose is lowered throughout life

Blood was sampled from a subset of mice at weaning prior to KD initiation (baseline levels) and approximately every 10 days thereafter to verify ketosis in KD-treated mice. Figure 2C,D depict the levels of BHB and glucose at three time points: at baseline (approximately P22), during treatment (P32–39), and the final time-point (the last measurement prior to death).

KD-treatment elevated blood BHB levels of *Kcna1*-null mice throughout life ($F_{2,39} = 100.5$, $p < 0.05$) to levels that were higher than both SD-treated epileptic and WT controls ($F_{2,39} = 110.5$, $p < 0.0001$) (Fig. 2C). BHB levels during treatment and at the final time-point were greater than pre-KD levels ($p < 0.001$ and $p < 0.001$, respectively) and were higher than in SD-treated *Kcna1*-null and WT controls ($p < 0.0001$ and $p < 0.0001$, respectively, Tukey's multiple comparisons post-hoc test). KD treatment reduced glucose levels in *Kcna1*-null mice when compared to SD controls ($F_{2,41} = 8.6$, $p < 0.001$) and to its own baseline levels ($p < 0.005$, Tukey's multiple comparisons post hoc test) (Fig. 2D). These data indicate that KD-treated *Kcna1*-null mice maintained ketosis and reduced blood glucose throughout life.

DISCUSSION

To our knowledge, the present study is the first to specifically assess longevity, a clinically relevant end-point, in an animal model of SUDEP. Herein, we report that the KD, an efficacious treatment for refractory and severe seizures, extends the lifespan of *Kcna1*-null mice.

This study begins to address two critical needs. First, there is a need to identify and expand clinically relevant animal models that express multiple risk factors and experience terminal events associated with human SUDEP.^{11–13} Previous studies have reported that *Kcna1*-null mice model multiple SUDEP risk factors and terminal events because they have (1) spontaneous seizures and (2) a severe seizure phenotype with myoclonic and generalized tonic–clonic (GTC) seizures (3) that are refractory to traditional antiseizure drugs (e.g., carbamazepine, (Simeone TA, Kim DY, Simeone KA, Rho JM) unpublished data). (4) *Kcna1*-null mice are cognitively impaired and (5) have cardiac arrhythmias; the terminal events associated with their sudden death include (6) a GTC seizure followed by (7) enhanced postictal parasympathetic-mediated bradyarrhythmias, and (8) asystole and death.^{5,9,10,17}

There is also a critical need to identify therapies that may increase the lifespans of patients at risk for SUDEP. As a basic science and clinical community we need to be

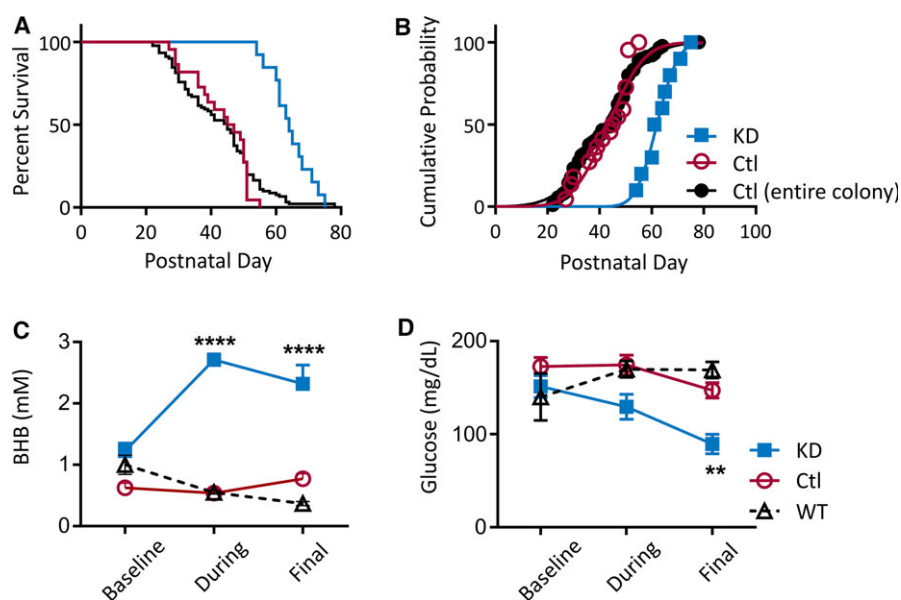


Figure 2.

KDiet increases longevity of *Kcna1*-null mice. **(A)** SD-treated control epileptic mice that were a part of this longevity study (Ctl; $n = 22$, red lines) exhibited the same survival curve as retrospective analyses of the *Kcna1*-null mice of the entire colony ($n = 90$, Ctl, entire colony, black lines). KD treatment significantly increased the survival of in *Kcna1*-null mice (KD; $n = 10$; $p < 0.0001$, blue lines). **(B)** Cumulative probability of death curves also demonstrated the rightward shift of the population as a whole in KD animals. Data points were fit with a cumulative Gaussian nonlinear regression and Goodness of Fit R^2 ranged from 0.91 to 0.98. **(C, D)** Measurements of blood BHB and glucose levels at baseline (approximately P22), during treatment (P32–39), and prior to death. Final measurements were taken between P35 and P49 for SD-treated *Kcna1*-null (Ctl, red) mice, between P50 and P70 for KD-treated *Kcna1*-null mice (KD, blue), and between P50 and P70 for SD-treated wild-type (WT, black) mice ($n = 5–9$ for each group at each time point), $**p < 0.001$; $****p < 0.0001$ versus baseline measurements.

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vigilant in terms of discussing specific details of how a drug may influence specific SUDEP features for each model. Previous studies have reported that the KD increases seizure threshold (latency to seizure onset) in multiple models of inducible seizures, including *Kcna1*+/- mice and the *Scn1a*+/- mouse line (a model of Dravet syndrome).^{18,19} Furthermore, here we found that the progression of the severe epilepsy phenotype was delayed in KD-treated mice. Our data support previous findings where KD treatment delayed disease progression in the pilocarpine kindling epilepsy model²⁰; however, lifespan was not assessed in this study.

KD has been shown to extend longevity in animal models of amyotrophic lateral sclerosis and succinic semialdehyde dehydrogenase deficiency.^{21,22} Our data further support the use of the KD as a means to improve longevity, as we found lifespan increased by approximately 47% in *Kcna1*-null mice. Further studies are required to determine whether KD-treated mice experience similar terminal events as SD-treated cohorts, including the final GTC and cardiorespiratory failure.

There is a rich history of dietary manipulations improving central and peripheral health and promoting longevity.²³ Much of our current knowledge comes from investigations into the effects of caloric restriction and protein restriction

on lifespan.^{23,24} Possible mechanisms include improvements to mitochondrial bioenergetics. Improving mitochondrial health, specifically increasing ATP-producing respiratory rates and reducing reactive species' post-translational impairment of complex I, reduces in vivo seizures and in vitro hippocampal hyperexcitability.^{5,6,15} Not surprisingly, the KD engages many pro-mitochondrial pathways.^{5,25–27} Furthermore, the KD affords neuroprotection and improves learning, memory, and synaptic integrity in *Kcna1*-null mice, potentially reducing another SUDEP risk factor.^{5,6,17} Ketone bodies, which are elevated by KD, improve synaptic functioning, reduce hyperexcitability in vitro and attenuate seizures in vivo, primarily by limiting mitochondrial permeability transition.^{5,17} KD-induced increase in polyunsaturated fatty acids are an endogenous ligand for Peroxisome proliferator-activated receptor gamma, a nutritionally activated transcription factor upstream of several pro-mitochondrial and anti-inflammatory genes. Activation of PPAR γ is associated with increased longevity.^{25,28} However, whether PPAR γ activation is involved in the antiseizure mechanism of action of the KD has yet to be determined.

In conclusion, the KD actions likely involve a myriad of effectors and mediators that dampen an equally complex interplay between seizures and SUDEP. Although, future

studies are needed to determine the mechanisms underlying the KD effects on longevity, these findings suggest the KD may be a possible treatment to improve longevity.

ACKNOWLEDGMENTS

The authors thank S. Iyer for her technical assistance. This work was supported by the National Institutes of Health R01 NS072179 (KAS), R01 NS085389 (TAS), Citizens United for Research in Epilepsy Award (TAS), and the Canadian Institutes of Health Research (JMR). This project was also supported by Grant Number G20RR024001 from the National Center for Research Resources. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Center for Research Resources or the National Institutes of Health. Please note, KA Simeone has published under the names K Dorenbos, KA Fenoglio, and KA Simeone.

DISCLOSURE

None of the authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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